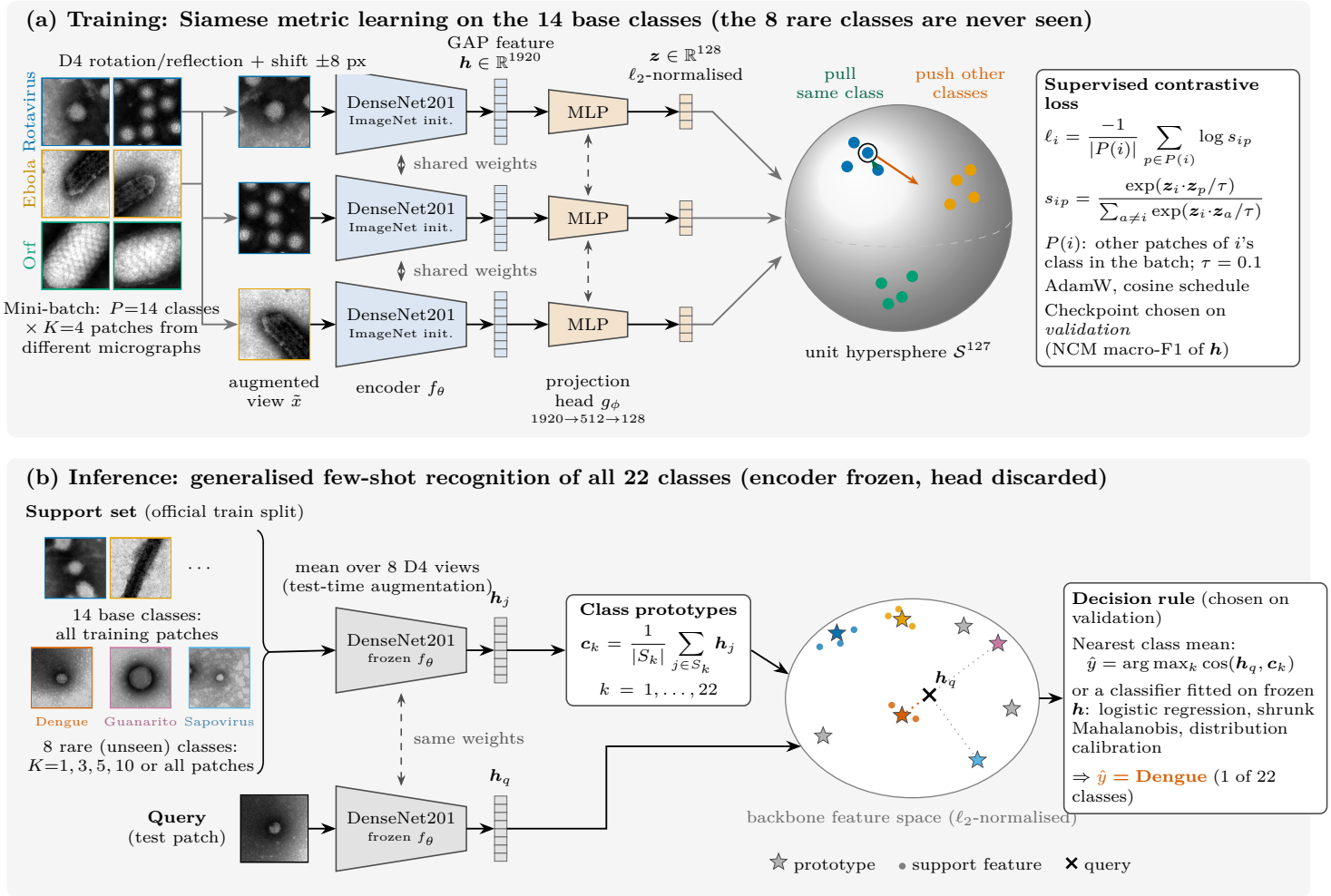

(a) Training: Siamese metric learning on the 14 base classes (the 8 rare classes are never seen)
D4 rotation/reflection + shift ±8 px
GAP feature
$\boldsymbol{h} \in \mathbb{R}^{1920}$
$\boldsymbol{z} \in \mathbb{R}^{128}$
ℓ_2-normalised
Ebola
Rotavirus
Orf
DenseNet201
ImageNet init.
shared weights
MLP
pull
same class
push other
classes
Supervised contrastive loss
$\ell_i = \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log s_{ip}$
$s_{ip} = \frac{\exp(\boldsymbol{z}_i \cdot \boldsymbol{z}_p / \tau)}{\sum_{a \neq i} \exp(\boldsymbol{z}_i \cdot \boldsymbol{z}_a / \tau)}$
$P(i)$: other patches of i's class in the batch; $\tau = 0.1$
AdamW, cosine schedule
Checkpoint chosen on validation
(NCM macro-F1 of $\boldsymbol{h}$)
Mini-batch: P=14 classes
× K=4 patches from
different micrographs
augmented
view $\tilde{x}$
encoder f_θ
projection
head g_ϕ
1920→512→128
unit hypersphere $\mathcal{S}^{127}$
(b) Inference: generalised few-shot recognition of all 22 classes (encoder frozen, head discarded)
Support set (official train split)
14 base classes:
all training patches
Dengue
Guanarito
Sapovirus
8 rare (unseen) classes:
K=1, 3, 5, 10 or all patches
mean over 8 D4 views
(test-time augmentation)
$\boldsymbol{h}_j$
DenseNet201
frozen f_θ
same weights
Class prototypes
$\boldsymbol{c}_k = \frac{1}{|S_k|} \sum_{j \in S_k} \boldsymbol{h}_j$
$k = 1, \ldots, 22$
Query
(test patch)
$\boldsymbol{h}_q$
backbone feature space (ℓ_2-normalised)
Decision rule (chosen on validation)
Nearest class mean:
$\hat{y} = \arg\max_k \cos(\boldsymbol{h}_q, \boldsymbol{c}_k)$
or a classifier fitted on frozen $\boldsymbol{h}$: logistic regression, shrunk Mahalanobis, distribution calibration
⇒ $\hat{y}$ = Dengue (1 of 22 classes)
prototype
support feature
query